

TD3 vs DDPG Performance Analysis in Autonomous Driving Environment

Structured research-style report built from deterministic evaluation summaries (20 episodes, 600-step horizon, exploration noise OFF), seed-aggregated metrics, grouped noise-level plots, algorithm comparisons, and summary text.

Project Title	TD3 vs DDPG Performance Analysis in Autonomous Driving Environment
Description	Comparison of reward systems across noise levels and algorithm behavior under deterministic evaluation.
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Summary Metrics

Evaluation metrics are reported as condition mean $\pm$ sample standard deviation (ddof=1) across independent training seeds. Crash rate represents deterministic collision frequency during evaluation, and convergence episode indicates sample-efficient training progression.

Experiment	TD3 Reward $\pm$ SD	DDPG Reward $\pm$ SD	TD3 Crash $\pm$ SD	DDPG Crash $\pm$ SD	TD3 Conv	DDPG Conv
R1_N1	30.00 $\pm$ 0.00	15.70 $\pm$ 11.71	0.00 $\pm$ 0.00%	33.33 $\pm$ 57.74%	30 $\pm$ 17	32 $\pm$ 20
R1_N2	28.00 $\pm$ 1.86	27.87 $\pm$ 3.70	0.00 $\pm$ 0.00%	15.00 $\pm$ 25.98%	32 $\pm$ 17	37 $\pm$ 15
R1_N3	21.32 $\pm$ 15.04	14.06 $\pm$ 13.97	0.00 $\pm$ 0.00%	5.00 $\pm$ 8.66%	38 $\pm$ 15	71 $\pm$ 64
R2_N1	119.23 $\pm$ 0.49	77.30 $\pm$ 65.75	0.00 $\pm$ 0.00%	66.67 $\pm$ 57.74%	28 $\pm$ 14	29 $\pm$ 15
R2_N2	118.95 $\pm$ 1.01	52.20 $\pm$ 43.61	0.00 $\pm$ 0.00%	11.67 $\pm$ 20.21%	29 $\pm$ 13	35 $\pm$ 21
R2_N3	121.63 $\pm$ 4.09	47.76 $\pm$ 44.22	0.00 $\pm$ 0.00%	51.67 $\pm$ 50.08%	37 $\pm$ 25	53 $\pm$ 38
R3_N1	117.55 $\pm$ 101.56	77.34 $\pm$ 25.81	0.00 $\pm$ 0.00%	66.67 $\pm$ 57.74%	28 $\pm$ 14	29 $\pm$ 15
R3_N2	163.68 $\pm$ 13.07	136.59 $\pm$ 24.11	0.00 $\pm$ 0.00%	28.33 $\pm$ 33.29%	35 $\pm$ 23	30 $\pm$ 13
R3_N3	163.72 $\pm$ 16.21	99.70 $\pm$ 85.93	3.33 $\pm$ 5.77%	21.67 $\pm$ 29.30%	30 $\pm$ 13	42 $\pm$ 21
R4_N1	258.69 $\pm$ 3.78	150.07 $\pm$ 129.58	0.00 $\pm$ 0.00%	66.67 $\pm$ 57.74%	28 $\pm$ 14	29 $\pm$ 15
R4_N2	219.14 $\pm$ 11.76	127.17 $\pm$ 30.77	15.00 $\pm$ 8.66%	76.67 $\pm$ 15.28%	29 $\pm$ 12	30 $\pm$ 13
R4_N3	205.37 $\pm$ 65.97	113.35 $\pm$ 104.92	31.67 $\pm$ 34.03%	40.00 $\pm$ 37.75%	44 $\pm$ 37	52 $\pm$ 34

Analysis by Noise Level

Each subsection fixes one noise level and compares R1-R4 for TD3 and DDPG using the grouped plots generated from seed-aggregated curves.

Noise Level N1

Grouped experiment curves compare R1-R4 within a fixed noise setting. Each plot shows the seed-aggregated mean with light uncertainty shading.



Summary

Noise N1: higher reward tends to appear in R4, while stronger stability tends to appear in R1. TD3 reward 131.37 ± 94.56 , DDPG reward 80.10 ± 54.95 ; TD3 crash $0.00 \pm 0.00\%$, DDPG crash $58.33 \pm 16.67\%$.

Key Takeaways

- Safety must be interpreted jointly with reward; high reward and high crash is unsafe.
- Lower standard deviation indicates more stable learning under this noise level.

Noise Level N2

Grouped experiment curves compare R1-R4 within a fixed noise setting. Each plot shows the seed-aggregated mean with light uncertainty shading.



Summary

Noise N2: higher reward tends to appear in R4, while stronger stability tends to appear in R1. TD3 reward 132.44 ± 80.79 , DDPG reward 85.96 ± 54.09 ; TD3 crash $3.75 \pm 7.50\%$, DDPG crash $32.92 \pm 30.04\%$.

Key Takeaways

- Safety must be interpreted jointly with reward; high reward and high crash is unsafe.
- Lower standard deviation indicates more stable learning under this noise level.

Noise Level N3

Grouped experiment curves compare R1-R4 within a fixed noise setting. Each plot shows the seed-aggregated mean with light uncertainty shading.



Summary

Noise N3: higher reward tends to appear in R4, while stronger stability tends to appear in R1. TD3 reward 128.01 ± 78.92 , DDPG reward 68.72 ± 46.11 ; TD3 crash $8.75 \pm 15.36\%$, DDPG crash $29.58 \pm 20.52\%$.

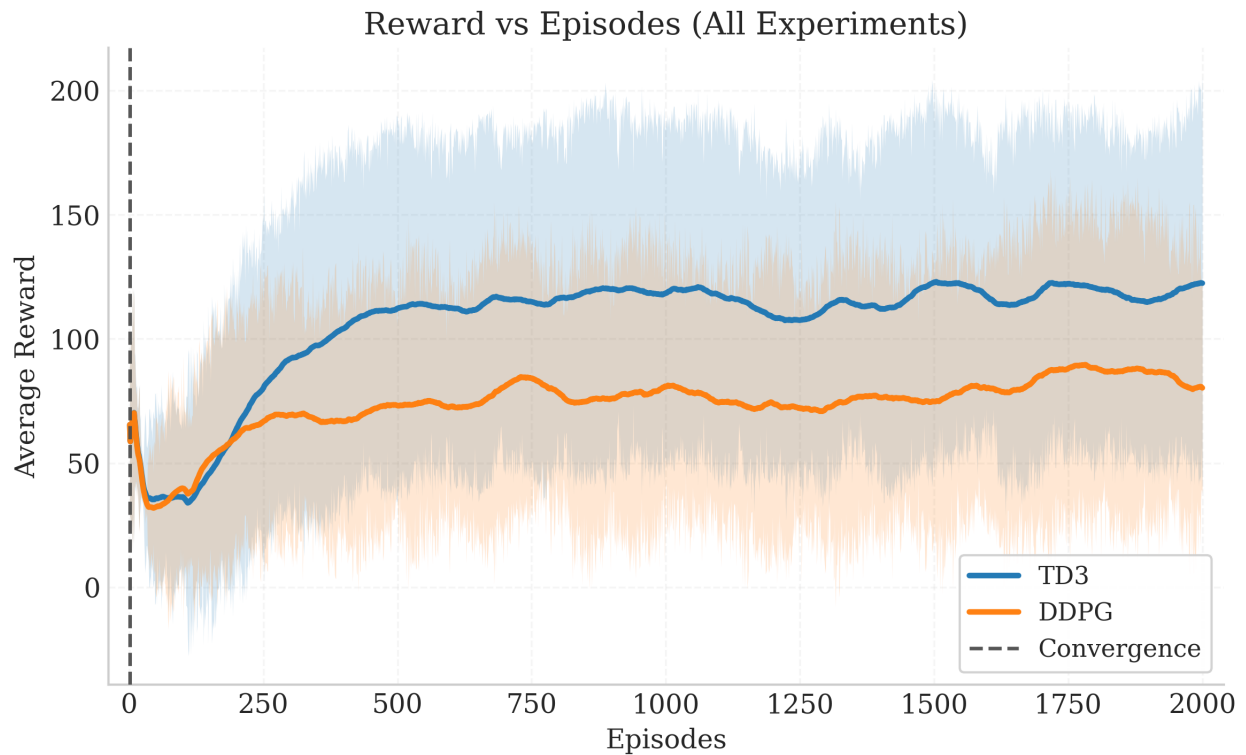
Key Takeaways

- Safety must be interpreted jointly with reward; high reward and high crash is unsafe.
- Lower standard deviation indicates more stable learning under this noise level.

Algorithm Comparison

The next pages compare TD3 and DDPG directly using the existing comparison plots, focusing on convergence speed, stability, and final performance.

TD3 vs DDPG Reward Comparison



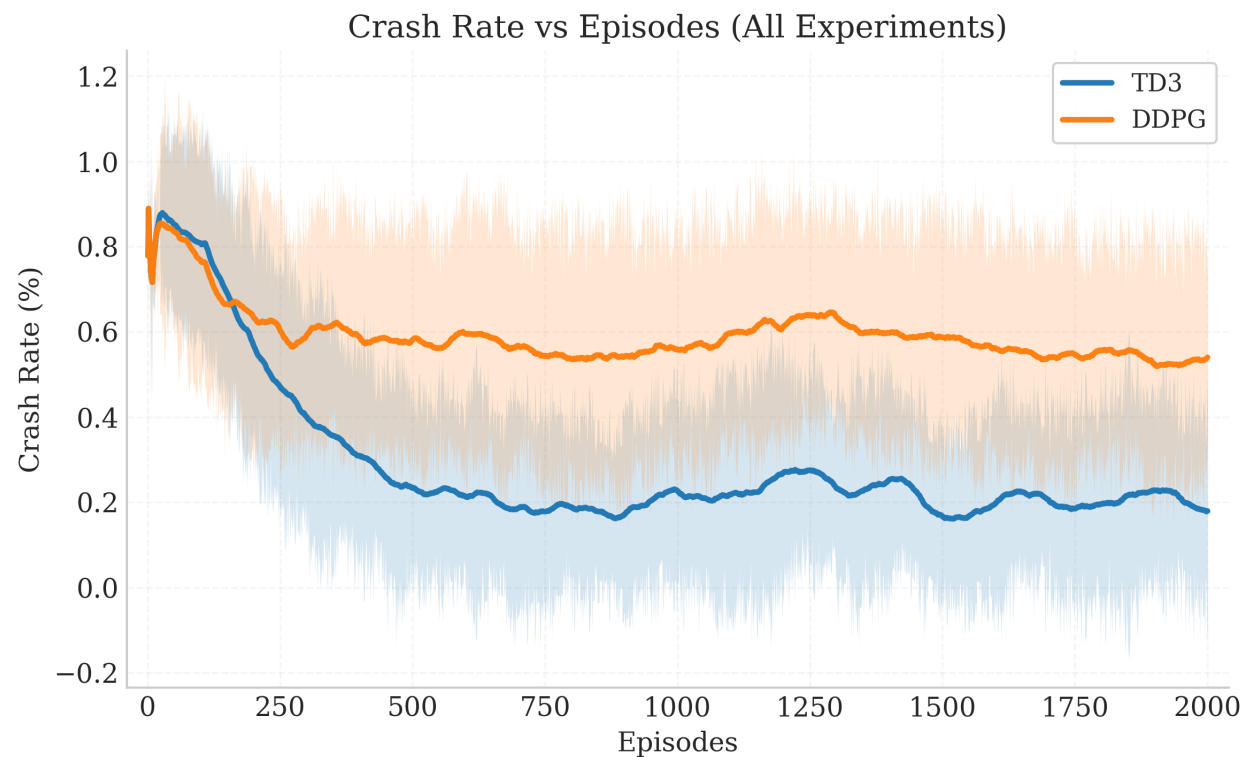
Caption. Reward vs Episodes across all experiments with uncertainty represented as mean $\pm$ standard deviation.

TD3 reward 130.61 ± 76.95 , DDPG reward 78.26 ± 47.51 ; TD3 crash $4.17 \pm 9.68\%$, DDPG crash $40.28 \pm 24.83\%$; TD3 convergence 32 ± 5 , DDPG convergence 39 ± 13 .

Key Takeaways

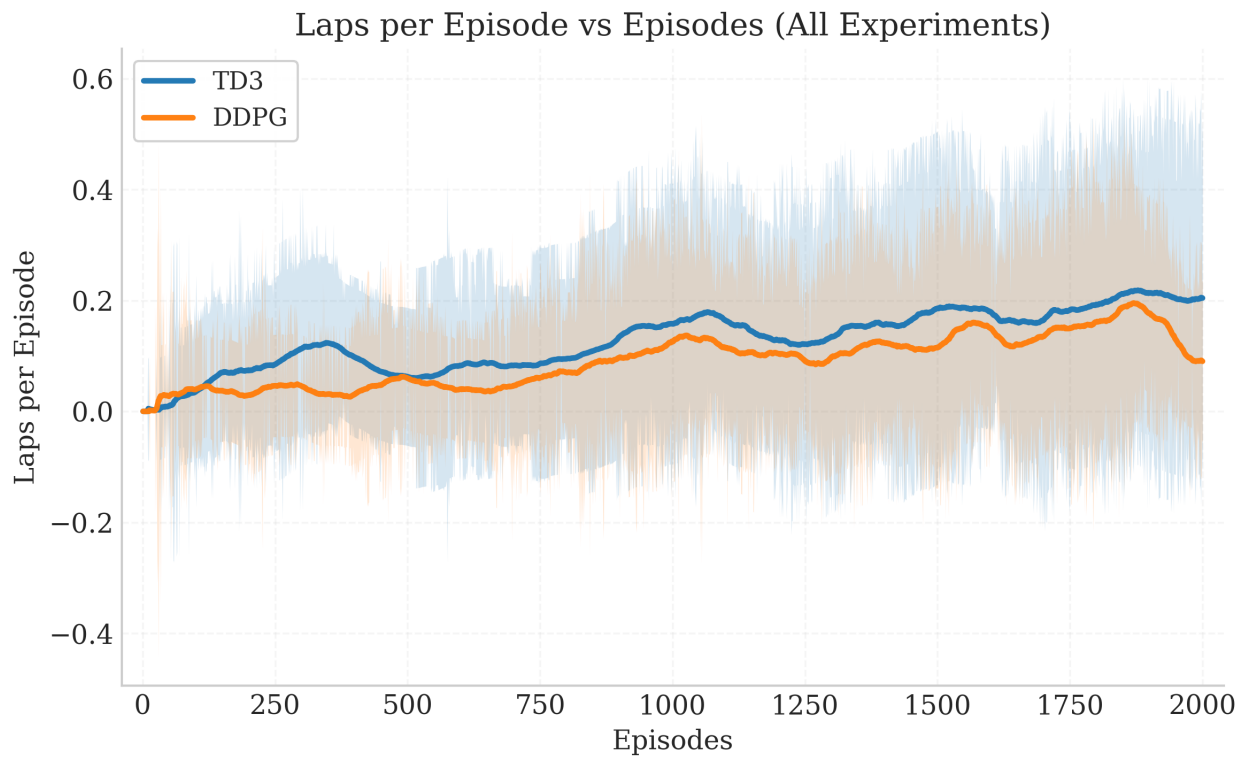
- Lower convergence episode implies faster sample-efficient learning.
- Reward must be interpreted with crash rate to avoid unsafe conclusions.

TD3 vs DDPG Crash Rate Comparison



Caption. Crash Rate vs Episodes across all experiments with uncertainty represented as mean $\pm$ standard deviation.

TD3 vs DDPG Laps Comparison



Caption. Laps vs Episodes across all experiments with uncertainty represented as mean $\pm$ standard deviation.

Key Aggregate Insights

The following section summarizes cross-noise trade-offs with explicit safety, stability, and sample-efficiency interpretation.

Highest reward-system trend across noise levels: R4. Most stable reward-system trend: R1. Lower crash tendency appears in TD3, while faster convergence appears in TD3. Noise generally reduces reward and increases variability, emphasizing a trade-off between raw performance and robustness.

Key Takeaways

- Stability-sensitive deployments should prioritize low variance and low crash behavior.
- Performance-oriented deployments should still verify safety under higher noise.

Final Conclusion

The results indicate a trade-off between performance and robustness. Highest reward trend is associated with R4, while greater stability trend is associated with R1. DDPG can achieve higher rewards and faster convergence in several settings, while TD3 can provide more stable learning and improved safety in certain conditions. Higher noise levels generally reduce reward and increase instability.

Key Takeaways

- TD3 is preferable when stability and safety are prioritized.
- DDPG is preferable when higher reward is prioritized and crash behavior remains acceptable.